

$$\begin{aligned}
 1) \quad G(s) &= \frac{10(s+3)}{s(s^2+s+2)} \quad \frac{\omega_n^2}{s^2+2\zeta\omega_n+\omega_n^2} \\
 &= 10\left(\frac{1}{s}\right)\left(\frac{3(s+3)}{3}\right)\left(\frac{1}{s^2+s+2}\right) \\
 &= 30\left(\frac{1}{s}\right)\left(\frac{s+3}{3}\right)\left(\frac{\sqrt{2}^2}{s^2+2\left(\frac{1}{2\sqrt{2}}\right)\sqrt{2}s+\sqrt{2}^2}\right)\left(\frac{1}{2}\right) \\
 &= 15\left(\frac{1}{s}\right)\left(\frac{s+3}{3}\right)\left(\frac{\sqrt{2}^2}{s^2+2\left(\frac{1}{2\sqrt{2}}\right)\sqrt{2}s+\sqrt{2}^2}\right)
 \end{aligned}$$

$$\begin{aligned}
 \omega_n &= \sqrt{2} \\
 \zeta &= \frac{1}{2\sqrt{2}}
 \end{aligned}$$

Corner frequencies: $0.1\sqrt{2}$, $\sqrt{2}$, $10\sqrt{2}$

$$\frac{1}{j\omega} = -\frac{1}{\omega}j = \frac{1}{\omega}e^{-\frac{\pi}{2}j}$$

$$G(j\omega) = \frac{j\omega+3}{3}$$

$$\text{When } \omega \ll 3, G(j\omega) = 1$$

$$\text{When } \omega \gg 3, G(j\omega) = \frac{j\omega}{3} = \frac{\omega}{3}e^{\frac{\pi}{2}j}$$

$$\text{When } \omega \ll \sqrt{2}, G(j\omega) = 1$$

$$\begin{aligned}
 \text{When } \omega \gg \sqrt{2}, G(j\omega) &= \frac{\sqrt{2}^2}{\omega^2}e^{j(-\pi)} \\
 &= \frac{2}{\omega^2}e^{j(-\pi)}
 \end{aligned}$$

Magnitude/dB

$\rightarrow \frac{1}{s}$

$\downarrow \frac{1}{\sqrt{2}}$

$$\frac{s^2 + 2\left(\frac{1}{\sqrt{2}}\right)\sqrt{2}s + \sqrt{2}^2}{\sqrt{2}^2}$$

$\rightarrow 20 \lg |s|$

$\rightarrow \frac{s+3}{3}$

$0.1\omega_n$

ω_n

$10\omega_n$

